



GeForce RTX 3060 Ti Aero ITX

MSI and Palit unveil most powerful graphics card for small form-factor systems, running on 4864 CUDA cores. <http://dgit.in/feb21-57>



Chunky GPU

Aorus has added a quadruple slot RTX 3090 Master with a beefy triple-fan heatsink; same specs. <http://dgit.in/feb21-58>

The lag that no one talks about

Agent001 gets into input lag

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e're not alien to the concept of lag. You've either experienced it personally or have heard it as an excuse

from one of your teammates after they've completely dropped the ball during an online match. Lag, ping, latency, turn around time, etc. are all different terms that have been used to mean the same thing but they're actually quite different. We aren't getting into the definitions of the above terms but simply put, lag is the phenomenon that you experience when there's a dissonance between your actions and the manifestations of those actions on screen. Perhaps that was a little too twisted of an explanation so let me try again. If your PC seems to be responding a little late to your inputs, then you're experiencing lag. A common example in video games is when you're tapping on the left mouse button and your weapon appears to fire a little later. This can be a very simple difference which works out to about 30-60 milliseconds or it could be as high as a second. The latter can make for a really frustrating experience.

When we talk about lag, we're usually dealing with network latency i.e. the time it takes for a data packet to be sent from your PC to arrive at a remote server and for the response to come back to your PC. Then there's

input lag, which is the time difference between clicking or tapping on a key and the corresponding action happening on screen. The impact of this input delay shouldn't be considered to be an addition to the network latency. The manner in which the game has been coded affects whether there would be an overlap between



the two. In multiplayer or competitive games, the developers focus on minimizing all forms of lag. Apex Legends has an average input lag of about 50 milliseconds on a mid-range graphics card that's connected to a monitor running at 60 Hz. As you bump up the refresh rate, the input lag goes down by a considerable amount.

Graphics card manufacturers have also incorporated new features that help in reducing input lag because their graphics cards are responsible for most of input latency. If we're being specific, we can break this down into input latency i.e. delay from your keyboard or mouse, then we have PC latency which has to do with the time taken for the signal to be processed within your PC and then finally you have the display latency which deals with the processing done by the monitor. All of these combined make up the system latency which we informally refer to as the input latency.

NVIDIA has something called REFLEX and AMD has Anti-Lag. Both work on reducing the system latency. And for reducing the display latency, you can simply use a monitor with a faster refresh rate. And lastly, you have the input lag i.e. what your mouse or keyboard contributes is very minimal and using devices from any respectable brand should help you in that regard.

Figuring out how much system latency you experience is an entirely different ball game altogether. You are trying to measure values that are as small as 15-to-80 milliseconds. That is usually tackled in laboratories with the use of oscilloscopes or high-speed cameras which record the input device i.e. mouse and the screen